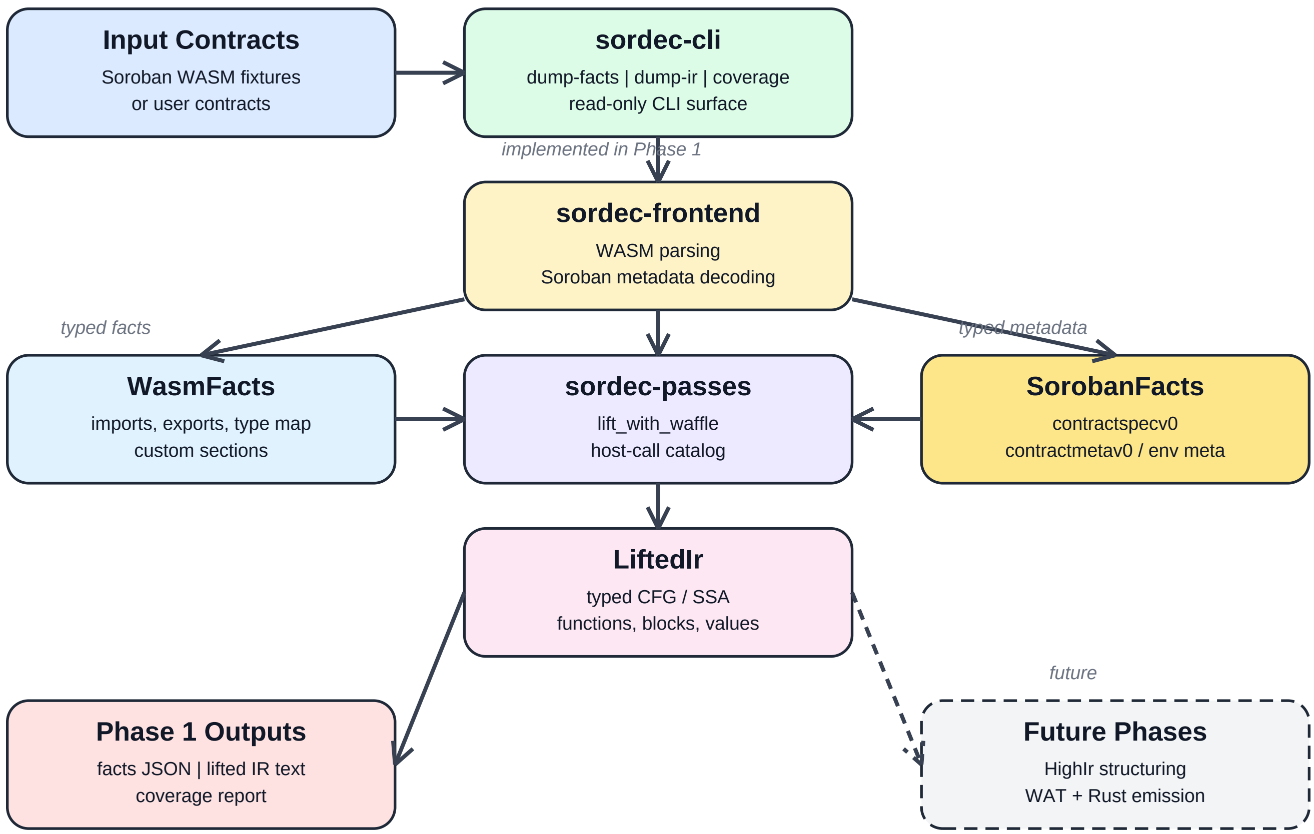
Sordec Phase 1 Architecture

A component-oriented view of what the repository implements today as the first phase of the Soroban reverse-engineering tool.

Phase 1 Component Flow



Phase 1 Scope

Phase 1 is the foundation and inspection layer of the broader decompiler project. It does not yet reconstruct source code. Instead, it turns raw Soroban WASM into typed facts, typed lifted IR, and measurable inspection outputs.

The three user-visible deliverables are `dump-facts`, `dump-ir`, and `coverage`.

Primary Components

- **sordec-cli**: read-only command surface that executes the implemented pipeline and renders facts, IR, and coverage.
- **sordec-frontend**: parses core WASM structure and decodes Soroban custom sections into typed metadata.
- **sordec-ir**: defines the durable typed boundaries `WasmFacts`, `LiftedIr`, and scaffolded `HighIr`.
- **sordec-passes**: wraps `waffle` for WASM-to-SSA/CFG lifting and ships the Soroban host-call catalog.
- **sordec-common**: shared IDs, arenas, diagnostics, provenance, and unknown-reason tracking.
- **sordec-driver**: future end-to-end orchestrator; in Phase 1 it intentionally stops at the wired front half.
- **sordec-backend**: placeholder for annotated WAT and compilable Rust emission in later phases.
- **samples/ and tools/**: reproducible real-contract corpus and verification scripts used to prove Phase 1 on real inputs.

Implemented Data Flow

A Soroban contract enters through the CLI, is parsed by the frontend into `WasmFacts` and optional `SorobanFacts`, then lifted by `sordec-passes::lift_with_waffle` into `LiftedIr`. The CLI finally renders either JSON facts, human-readable IR, or a coverage summary.

What Is Deliberately Deferred

Semantic pattern recovery, control-flow structuring, `HighIr` lowering, annotated WAT emission, and recovered Rust are all later-phase work. Phase 1 exists to make those later phases possible on top of a verified typed foundation.